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Surgical instrument having a self-centered quick connect interface

Abstract

A quick connect tool, a replaceable tool, and methods are disclosed. The replaceable tool includes a shaft having a tool end, a coupling end opposite the tool end, and a longitudinal axis extending between the coupling end and the tool end. The shaft has a tool portion defining the tool end, a coupling portion defining the coupling end, the coupling portion having an outer peripheral surface shaped into a flat portion coextensive with an arcuate portion cooperatively forming a D-shaped cross-section. The flat portion and the arcuate portion extend along the longitudinal axis. The outer peripheral surface defines a peripheral groove extending transversely with respect to the longitudinal axis. The peripheral groove is configured to receive a bearing, and the outer peripheral surface also defines a bearing surface having a shape conforming to a partial spiral extending around the longitudinal axis and located at the coupling end of the shaft.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) Not Applicable.

STATEMENT AS TO RIGHTS TO INVENTIONS MADE UNDER FEDERALLY SPONSORED RESEARCH AND DEVELOPMENT

(2) Not applicable.

BACKGROUND ART

(3) Quick connect interfaces (or simply “quick connects”) are widely used in surgical instruments to connect a replaceable tool, such as a bit, to a drive mechanism, a ratcheting or non-ratcheting handle, or a motor-operated drive. These surgical instruments may be used to perform a variety of surgical tasks, including drilling, reaming, tapping, placement of bone screws, assembly of spinal constructs, and the like. One common type of quick connect interface that may be employed is an Association for Osteosynthesis (AO)-style quick connect interface, typically referred to simply as an “AO quick connect interface.” Such an interface may be used with a wide variety of different types of replaceable tools, as well as different drive mechanisms.

(4) A typical AO quick connect interface includes a body member configured to receive a coupling end of a shaft having a D-shaped profile. The D-shaped profile defines a flat portion configured to extend past and engage another flat portion (referred to herein as the “AO flat”) formed in an inner cavity of the body member of the AO quick connect interface. This engagement prevents the rotation of the shaft with respect to the body member. The conventional AO quick connect interface also includes one or two openings in the body member, and one or two ball bearings. Each ball bearing is positioned within one of the one or two openings, and a sleeve surrounds the body member. The shaft may be retained in the body member by the operation of the ball bearing. The opening extends from an exterior of the body member to the inner cavity within the body member. The ball bearing can be positioned partially within the inner cavity and engage a groove formed in the end of the shaft of the replaceable tool. The ball bearing is held in this engaged position by a substantially straight-walled portion of an interior of the sleeve.

(5) One of the challenges of the conventional AO coupling is for a user to align the flat portion on the coupling end of the replaceable tool with the flat portion forming a boundary of the inner cavity within the body member. The conventional AO Coupling requires the user to partially insert the coupling end of the replaceable tool into the inner cavity, and then rotate the replaceable tool relative to the body member of the quick connect coupling while forcing the replaceable tool into the inner cavity until the flat portion of the replaceable tool is aligned with the flat portion in the inner cavity. Once the flat portion of the replaceable tool is aligned with the flat portion of the inner cavity, the user can move the coupling end of the replaceable tool further into the inner cavity, thereby seating the coupling end of the replaceable tool into the inner cavity. The replaceable tool and the conventional AO coupling lack any mechanism constructed to guide the flat portions of the replaceable tool and the inner cavity into alignment. This causes surgeons and their assistants to fiddle with the replaceable tool and the AO coupling attached to the selected drive mechanism in the middle of surgery, in order to seat the coupling end of the replaceable tool into the inner cavity.

SUMMARY OF THE INVENTION

(6) To this end, a need exists for an improved coupling assembly, conforming to standards set by the Association for Osteosynthesis, that includes a replaceable tool and a quick connect interface constructed to guide the flat areas on the replaceable tool and within the inner cavity into alignment, to more easily seat the coupling end of the replaceable tool into a bore of a body member of the quick connect interface. It is to such an improved coupling assembly that the inventive concepts disclosed herein are directed. In one aspect, the problem of seating the

replaceable tool into the quick connect interface is addressed by forming a bearing surface on a coupling end of the replaceable tool and an alignment surface on an alignment member within a bore of the quick connect interface such that that when a coupling end of the replaceable tool is positioned within the bore, the bearing surface of the replaceable tool engages the alignment surface and causes rotary motion of the replaceable tool so as to align the coupling end of the replaceable tool with the bore.

(7) In some embodiments, the bearing surface has a partial spiral shape and is positioned on a coupling portion having a D-shaped cross-section of a shaft of the replaceable tool. The shaft has a tool end opposite to a coupling end, and a longitudinal axis extending from the tool end to the coupling end. The coupling portion extends from the coupling end towards the tool end.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate one or more implementations described herein and, together with the description, explain these implementations. The drawings are not intended to be drawn to scale, and certain features and certain views of the figures may be shown exaggerated, to scale or in schematic in the interest of clarity and conciseness. Not every component may be labeled in every drawing. Like reference numerals in the figures may represent and refer to the same or similar element or function. In the drawings:

(2) FIG. 1 is a side elevational view of a surgical instrument that includes a coupling assembly having a replaceable tool and a quick connect interface constructed in accordance with the present disclosure attached to a drive mechanism.

(3) FIG. 2A is a perspective view of a coupling portion of a shaft of the replaceable tool of FIG. 1.

(4) FIG. 2B is an end view of the coupling portion of the shaft of the replaceable tool of FIG. 1.

(5) FIG. 3A is a cross-sectional side view of the quick connect interface of FIG. 1 showing, among other things, an alignment member within a bore of a socket portion of the quick connect interface, a ball bearing and a coupling sleeve in an unlocked position.

(6) FIG. 3B is a cross-sectional side view of the quick connect interface of FIG. 1 showing, among other things, the alignment member within the bore of the socket portion of the quick connect interface, the ball bearing and the coupling sleeve in a locked position.

(7) FIG. 4 is a top plan view of the alignment member constructed in accordance with the present disclosure.

(8) FIG. 5 is a front view of the quick connect interface showing relative dimensions of the bore within the socket portion and the alignment member.

(9) FIG. 6A is a perspective view of a body member of the quick connect interface showing, among other things, a linear groove and pin slot.

(10) FIG. 6B is another cross-sectional side view of the quick connect interface of FIG. 1 showing, among other things, the linear groove and the pin slot;

(11) FIG. 7 is a cross-sectional front view of the quick connect interface of FIG. 1 showing, among other things, the sleeve in an unlocked position with an inner surface of the coupling sleeve spaced a distance away from the ball bearing.

DETAILED DESCRIPTION

(12) The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

(13) As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to

only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by anyone of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

(14) In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the inventive concept. This description should be read to include one or more and the singular also includes the plural unless it is obvious that it is meant otherwise.

(15) Further, use of the term “plurality” is meant to convey “more than one” unless expressly stated to the contrary.

(16) As used herein, qualifiers like “substantially,” “about,” “approximately,” and combinations and variations thereof, are intended to include not only the exact amount or value that they qualify, but also some slight deviations therefrom, which may be due to manufacturing tolerances, measurement error, wear and tear, stresses exerted on various parts, and combinations thereof, for example.

(17) The use of the term “at least one” or “one or more” will be understood to include one as well as any quantity more than one. In addition, the use of the phrase “at least one of X, V, and Z” will be understood to include X alone, V alone, and Z alone, as well as any combination of X, V, and Z.

(18) The use of ordinal number terminology (i.e., “first”, “second”, “third”, “fourth”, etc.) is solely for the purpose of differentiating between two or more items and, unless explicitly stated otherwise, is not meant to imply any sequence or order or importance to one item over another or any order of addition.

(19) As used herein, all numerical values or ranges include fractions of the values and integers within such ranges and fractions of the integers within such ranges unless the context clearly indicates otherwise. Thus, to illustrate, reference to a numerical range, such as 1-10 includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, as well as 1.1, 1.2, 1.3, 1.4, 1.5, etc., and so forth. Reference to a range of 1-50 therefore includes 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, etc., up to and including 50. Similarly, fractional amounts between any two consecutive integers are intended to be included herein, such as, but not limited to, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, and 0.95. For example, the range 3 to 4 includes, but is not limited to, 3.05, 3.1, 3.15, 3.2, 3.25, 3.3, 3.35, 3.4, 3.45, 3.5, 3.55, 3.6, 3.65, 3.7, 3.75, 3.8, 3.85, 3.9, and 3.95. Thus, even if specific data points within the range, or even no data points within the range, are explicitly identified or specifically referred to, it is to be understood that any data points within the range are to be considered to have been specified, and that the inventor possessed knowledge of the entire range and the points within the range.

(20) Finally, as used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment.

(21) Referring now to the figures, and in particular to FIG. **1**, shown therein is a surgical instrument **10** including a drive mechanism **12** attached to a coupling assembly **13**. The coupling assembly **13** includes a quick connect interface **14**, and a replaceable tool **16**. The quick connect interface **14** As will be described in more detail below, the quick connect interface **14** and the replaceable tool **16** of the surgical instrument **10** are configured to provide a self-centering function when the replaceable tool **16** is placed within the quick connect interface **14**. Other than the components providing the self-centering function described below, the quick connect interface **14** may be constructed to conform to the requirements of an Association for Osteosynthesis (AO)-style quick connect interface, typically referred to simply as an “AO quick connect interface.”

(22) The components of the surgical instrument **10** will now be described in detail. As shown in

FIG. 1, the drive mechanism **12** is coupled to the quick connect interface **14**. In this example, the drive mechanism **12** is a non-ratcheting molded handle. Alternatively, the drive mechanism **12** may be implemented in other ways. For example, the drive mechanism **12** may include ratcheting and non-ratcheting handles or a motor-operated drive (such as a battery-powered drive). The surgical instrument **10** may include a longitudinal axis **18** extending through a center of the drive mechanism **12**, the quick connect interface **14**, and the replaceable tool **16** of the surgical instrument **10**. The longitudinal axis **18** may extend in both a first direction **20**, and a second direction **22**. The replaceable tool **16** (in this example, a star drive bit) is configured to be removably coupled to the quick connect interface **14** to permit replacement of the replaceable tool **16**.

(23) The drive mechanism **12**, as shown in FIG. 1, is provided with a first end **30**, a second end **32**, and a handle portion **34** extending between the first end **30** and the second end **32**. In the example shown, the handle portion **34** is configured to be gripped by an individual. In this regard, the handle portion **34** may be provided with a substantially cylindrical shaped cross-section, hex-shaped cross-section, or octagonal shaped cross-section to facilitate the handle portion **34** being securely gripped by the individual. To receive and securely receive the quick connect interface **14**, the handle portion **34** may be provided with a bore **40** extending through the second end **32** towards the first end **30**. The bore **40** may be aligned with the longitudinal axis **18** as shown in FIG. 1. It should be understood that the bore **40** may extend sufficiently into the handle portion **34** so as to securely receive a portion of the quick connect interface **14** therein. It should also be understood that the bore **40** may be shaped so as to receive the portion of the quick connect interface **14** without permitting rotation of the portion relative to the handle portion **34**. For example, the bore **40** may be provided with a square shaped cross-section, or a hex shaped cross-section for example.

(24) The replaceable tool **16**, as shown in FIG. 1, is provided with a shaft **50** having a tool end **52**, and a coupling end **54**. The shaft **50** is provided with a tool portion **56**, a coupling portion **58**, and an intermediate portion **60** extending between the tool portion **56** and the coupling portion **58**. The tool portion **56**, the coupling portion **58**, and the intermediate portion **60** can all be integrally formed as a single unit. In other embodiments, the tool portion **56**, the coupling portion **58**, and the intermediate portion **60** can be separate elements that are connected together.

(25) As shown in FIG. 1, the tool portion **56** is located at the tool end **54** of the shaft **50**. Similarly, the coupling portion **58** is located at the coupling end **52** of the shaft **50**. The tool portion **56**, the coupling portion **58** and the intermediate portion **60** may all extend on the longitudinal axis **18**. Also, the coupling end **52** may be opposite to the tool end **56**.

(26) Referring to FIG. 1 in combination with FIG. 2A, the coupling portion **58** is provided with an outer peripheral surface **70** shaped into a flat portion **72** coextensive with an arcuate portion **74** cooperatively forming a D-shaped cross-section. The flat portion **72** and the arcuate portion **74** extend along at least a portion of the longitudinal axis **18** from the tool end **52** towards the intermediate portion **60**. The outer peripheral surface **70** also defines a peripheral groove **76** extending transversely with respect to the longitudinal axis **18**.

(27) As will be discussed in more detail below, the quick connect interface **14** is provided with a bearing **80** (see FIG. 3A). The bearing **80** may be a ball bearing, for example. The peripheral groove **76** in some embodiments is configured to receive at least a portion of the bearing **80** to maintain the coupling portion **58** within the quick connect interface **14**. The bearing **80** may conform to the requirements of the Association for Osteosynthesis-style quick connect interface. Likewise, the flat portion **72**, the arcuate portion **74**, and the peripheral groove **76** may also conform to the requirements of the Association for Osteosynthesis style quick connect interface.

(28) The outer peripheral surface **70** of the coupling portion **58** also defines at least one, and preferably two bearing surface(s) **90** with each bearing surface **90** having a shape conforming to a partial spiral extending around the longitudinal axis **18** and located at the coupling end **52** of the shaft **50**.

(29) In the example shown, the at least one bearing surface **90** includes a first bearing surface **90a** and a second bearing surface **90b**. The first bearing surface **90a** extends from the coupling end **52** in a counterclockwise direction around the longitudinal axis **18** and towards the intermediate portion **60**. And, the second bearing surface **90b** extends in a clockwise direction around the longitudinal axis **18** and towards the intermediate portion **60**. Otherwise, the first bearing surface **90a** and the second bearing surface **90b** are identical in construction and function. For purposes of brevity, the first bearing surface **90a** will be described in detail hereinafter. And such description, is equally applicable to the second bearing surface **90b**.

(30) The first bearing surface **90a** has a first end **96** and a second end **98** opposed from the first end **96**. In the example shown, the first end **96** originates at the coupling end **52** of the shaft **50** and the second end **98** terminates at the flat portion **72** of the outer peripheral surface **70** of the shaft **50**. In some embodiments, should be understood that the first end **96** may originate a distance from the coupling end **52** of the shaft **50**. The first bearing surface **90a** extends at an angle between 30° and 150° relative to the longitudinal axis **18** of the shaft **50** of the replaceable tool **16**.

(31) As best shown in FIGS. 2A and 2B, the first bearing surface **90a** and the second bearing surface **90b** meet at a meeting location **100**, which may form an apex at the intersection of the first bearing surface **90a** and the second bearing surface **90b**. The first bearing surface **90a** and the second bearing surface **90b** may meet within a distance of 0 mm to 2 mm from the coupling end **52** of the shaft **50**.

(32) As best shown in FIG. 2B, the meeting location **100** may be linear and extend along a transverse axis **106**. The transverse axis **106** may be normal to the flat portion **72**. In some embodiments, the meeting location **100** may be offset radially from the transverse axis **106**. For example, a first axis **110** and a second axis **112** may intersect the transverse axis **106**. The first axis **110** may be radially offset an angle *a* within a range of about 0 degrees to about 30 degrees from the transverse axis **106**. The second axis **112** may be radially offset an angle *b* within the range of about 0 degrees to about -30 degrees from the transverse axis **106**. The meeting location **100** may be positioned between the first axis **110** and the second axis **112**.

(33) Referring again to FIG. 2A, the flat portion **72** has a first end **120** and a second end **122**. The first end **120** may be located at the coupling end **52** and the second end **122** may be located towards the intermediate portion **60** from the first end **120**. In some embodiments, the second end **122** of the flat portion **72** is located a first distance **124** from the coupling end **52**. The meeting location **100** may be within a second distance **126** from the coupling end **52**. In general, the second distance **126** is less than the first distance **124**. In some embodiments, the second distance **126** can be within a range of zero to 2 mm.

(34) Referring now to FIGS. 3A and 3B, the quick connect interface **14** will be described. The quick connect interface **14** includes a body member **140**, the bearing **80**, a bias assembly **142**, and a coupling sleeve **144**. The sleeve **144** extends around at least a portion of the body member **140** and is slidably movable on the body member **140**. In some embodiments, the coupling sleeve **144** surrounds the body member **140** in a concentric relationship. As will be described in more detail below, the body member **140**, the bearing **80**, the bias assembly **142**, and the coupling sleeve **144** of the quick connect interface **14** are configured to receive and secure the coupling portion **58** of the replaceable tool **16** within the body member **140** of the quick connect interface **14**. As previously mentioned, other than the components providing the self-centering function described in the present disclosure, the quick connect interface **14** may be constructed to conform to the requirements of an Association for Osteosynthesis (AO)-style quick connect interface, typically referred to simply as an “AO quick connect interface.”

(35) The body member **140** is provided with a socket portion **150** and shaft member **152**. The body member **140** includes a first end **154** and a second end **156** opposite the first end **154**. The longitudinal axis **18** extends through the body member **140** from the first end **154** to the second end **156**. The socket portion **150** of the body member **140** extends from the first end **154** towards the

second end **156**. The shaft member **152** of the body member **140** extends from the socket portion **150** to the second end **156**. The shaft member **152** is the portion of the quick connect interface **14** that may extend into the bore **40** of the drive mechanism **12** to secure the quick connect interface **14** to the handle portion **34** of the drive mechanism **12**. It should also be understood that the bore **40** and the shaft member **152** may be shaped so that the bore **40** may receive the shaft member **152** without permitting rotation of the shaft member **152** relative to the handle portion **34**.

(36) The socket portion **150** includes an inner surface **160** defining an opening **162** at the first end **154**, and defines a bore **166** extending from the opening **162** at the first end **154** towards the second end **156**. The inner surface **160** is shaped to receive the coupling portion **58** of the replaceable tool **16**. As best shown in FIG. 5, the inner surface **160** has a flat portion **170** and an arcuate portion **172** define a second D-shaped cross-section corresponding to the first D-shaped cross-section of the replaceable tool **16**. The flat portion **170** and the arcuate portion **172** of the inner surface **160** may be sized to conform to requirements of the Association for Osteosynthesis (AO)-style quick connect interface.

(37) The socket portion **150** of the body member **140** has an alignment member **180** within the bore **166**. In some embodiments, the socket portion **150** of the body member **140** may be shaped to form the alignment member **180**. In other embodiments, the alignment member **180** may be constructed separately and fixed to the inner surface **160** via welding, or one or more suitable fastener. The alignment member **180** has an outer surface **184** defining at least a portion of the flat portion **170** of the inner surface **160** of the socket portion **150**. The alignment member **180** is positioned adjacent to and/or within the bore **166** a distance **191** from the first end **154**. The alignment member **180** has a top surface **190** and a front surface **192** facing the first end **154** of the body member **140**. The top surface **190** of the alignment member **180** defines at least a portion of the flat portion **170** of the inner surface **160**. The front surface **192** has at least one alignment surface **200**. In the example shown, the front surface **192** has a first alignment surface **200a** and a second alignment surface **200b**. The first alignment surface **200a** extends at an angle **204** between about 30 degrees and about 60 degrees relative to the longitudinal axis **18**. The second alignment surface **200b** extends at an angle **206** between about 30 degrees and about 60 degrees relative to the longitudinal axis **18**. In some embodiments, the first alignment surface **200a** and the second alignment surface **200b** form a V-shape.

(38) As shown in FIG. 5, the bore **166** has a diameter **210** and the alignment member **180** has a height **212**. The height **212** of the alignment member **180** is less than the diameter **210** of the bore **166** so as to allow for the arcuate portion **74** of the outer peripheral surface **70** of the shaft **50** to enter and exit the bore **166**. In some embodiments, the height **212** of the alignment member **180** is in a range from 10% to 25% of the diameter **210** of the bore **166**.

(39) The first alignment surface **200a** and the second alignment surface **200b** of the alignment member **180** have a shape that upon interaction with one of the first bearing surface **90a** and the second bearing surface **90b** of the replaceable tool **16** causes the replaceable tool **16** to rotate about the longitudinal axis **18** thereby guiding the flat portion **72** (which may be referred to herein as a “first flat portion”) of the replaceable tool **16** into alignment with the flat portion **170** (which may be referred to herein as a “second flat portion”) of the alignment member **180**.

(40) As best shown in FIG. 4, the first alignment surface **200a** and the second alignment surface **200b** of the alignment member **180** may be planar and may be formed at non-normal angles relative to the longitudinal axis **18**.

(41) The bearing surface(s) **90** of the replaceable tool **16** and the alignment surface(s) **200a** and **200b** of the alignment member **180** are configured such that when the bearing surface(s) **90** are moved toward the alignment surface(s) **200a** and **200b** in a first direction **220** (see FIG. 1) at a first rate, the bearing surface(s) **90** and the alignment surface(s) **200a** and **200b** are engaged. Further linear motion of the bearing surface(s) **90** in the first direction **220** causes one of the bearing surfaces **90** to ride on one of the first alignment surface or the second alignment surface **200b**

converting the further linear motion of the bearing surface(s) **90** into a combination of (a) the further linear motion in the first direction **220** and (b) rotational motion. Thus, the bearing surface(s) **90** on the coupling portion **58** of the replaceable tool **16** and the alignment surfaces **200a** and **200b** of the alignment member **180** of the quick connect interface **14** are operable to translate linear motion into a combination of linear motion and rotational motion that serves to permit the user to insert the coupling portion **58** of the replaceable tool **16** into the bore **166** of the socket portion **150** of the quick connect interface **14** with a linear motion and automatically align the first D-shaped cross-section of the coupling portion **58** with the second D-shaped cross-section of the bore **166** of the socket portion **150** of the quick connect interface **14**.

(42) Referring to FIGS. **6A** and **6B**, other components of the quick connect interface **14** will be described hereinafter. The socket portion **150** of the body member **140** may be provided with a linear groove **230** positioned in an outer surface **232** of the socket portion **141** extending from a rear end **234** of the socket portion **150** towards the first end **154**.

(43) The coupling sleeve **144** surrounds the socket portion **150** of the body member **140** and is slidably movable on the body member **140**. The coupling sleeve **144** includes a first portion **240** and a second portion **242** with the first portion **240** positioned closer to the outer surface **232** than the second portion **242**. The first portion **240** is shaped so as to urge the bearing **80** to partially extend through a hole **246** into the bore **166** of the socket portion **150**. When the bearing **80** is maintained in a partially extended position through the hole **246** into the bore **166** of the socket portion **150**, and the coupling portion **58** of the replaceable tool **16** is positioned within the bore **166**, the bearing **80** extends into the peripheral groove **76** and functions to maintain the coupling portion **58** within the bore **166**. The socket portion **150** includes a lip **248** extending around a circumference of the hole **246** at the inner surface **160**. The bearing **80** has a diameter greater than the smallest diameter of the lip **248** at the inner surface **160** so that the bearing **80** may only partially cross the inner surface **160** of the socket portion **150**. The second portion **242** is shaped so as to not urge the bearing **80** into the bore **166**. When the second portion **242** overlaps the bearing **80**, the bearing **80** may freely move outside of the bore **166** when force is applied to the bearing **80**. Thus, when the coupling sleeve **144** is moved such that the second portion **242** overlaps the bearing **80**, the coupling portion **58** of the replaceable tool **16** can be moved into the bore **166** and/or removed from the bore **166** without substantial interference from the bearing **80**.

(44) In some embodiments, the coupling sleeve **144** includes an inner surface **250** positioned adjacent to the outer surface **232** of the socket portion **150** of the body member **140**. To guide the coupling sleeve **144** on the outer surface **232** of the socket portion **150**, the coupling sleeve **144** may include a pin member **254** connected to the coupling sleeve **144** and extending from the inner surface **250** and into the linear groove **230**. The linear groove **230** guides the pin member **254** in a linear direction as the coupling sleeve **144** is moved between a locked position in which the first portion **240** overlaps the bearing **80** and an unlocked position in which the second portion **242** overlaps the bearing **80**.

(45) The dimensions of the linear groove **230** may be compatible with the pin member **254**. The engagement of the pin member **254** within the linear groove **230** removes a rotational degree of freedom of the coupling sleeve **144** with respect to the body member **140**. The engagement of the pin member **254** within the linear groove **230** enables the coupling sleeve **144** to slide along the longitudinally axis **18** from the locked position to the unlocked position, provided a force is applied by the user to the coupling sleeve **144** toward the first end **154** that is sufficient to overcome the force of the bias assembly **142**. When the force applied by the user is removed from the coupling sleeve **144**, the bias assembly **142** applies a force to the coupling sleeve **144** to move the coupling sleeve toward the second end **156** thereby restoring the coupling sleeve **144** to the locked position.

(46) As previously mentioned, the quick connect interface **14** is provided with the bearing **80**. The peripheral groove **76** in some embodiments is configured to receive at least a portion of the bearing **80** to maintain the coupling portion **58** of the replaceable tool **16** within the bore of the quick

connect interface **14**. The bearing **80** is housed within the hole **246**. The hole **246** may be positioned so that the bearing **80** is in substantial alignment with the peripheral groove **76** in the coupling portion **58**.

(47) The bias assembly **142**, as shown in FIG. 3A, may surround a portion of the shaft member **152**, and may urge the coupling sleeve **144** toward the second end **156** of the body member **140**, so that, when at steady state, the coupling sleeve **144** is in the locked position. In one embodiment the bias assembly **142** is a compression spring that engages with the rear end **234** of the socket portion **150** of the body member **140** and the inner surface **250** of the coupling sleeve **144**. In use, a user may pull the coupling sleeve **144** toward the first end **154**, to the unlocked position, against the resistance of the bias assembly **142**.

(48) The coupling sleeve **144** may surround portions of the socket portion **150** and the shaft member **152**, the bearing **80**, and other components of the quick connect interface **14**. The coupling sleeve **144** may slide from the unlocked position, where the coupling sleeve **144** is shifted towards the first end **154** of the socket portion **150**, to a locked position, where the coupling sleeve **144** is shifted towards the second end **156** of the body member **140**.

(49) The coupling sleeve **144** slides from an unlocked position (FIG. 3A), where the sleeve **144** is shifted towards the first end **154** of the body member **140** of the quick connect interface **14**, to a locked position (FIG. 3B), where the coupling sleeve **144** is shifted towards the second end **156** of the body member **140**. The bias assembly **142**, e.g., a compression spring, may surround the shaft member **152**, and may urge the coupling sleeve **144** towards the second end **156**, so that, when at rest, the coupling sleeve **144** is in the locked position. Referring to FIGS. 3A and 3B-4, the coupling sleeve **144** may also include a camming ramp **280** formed in the interior surface **250** of the coupling sleeve **144**, proximate to the bearing **80**. The camming ramp **280** functions to urge the bearing **80** into the hole **246** as the coupling sleeve **144** is slid from the unlocked position (FIG. 3A) to the locked position (FIG. 3B).

(50) When the coupling sleeve **144** is returned to the unlocked position (FIG. 3A), pressure on the bearing **80** from the camming ramp **280** may be removed. The bearing **80** is allowed to recess in the hole **246** from the bore **166**. The coupling portion **58** of the replaceable tool **16** may then be removed from the quick connect interface **14**.

(51) To use the coupling assembly **13**, the coupling portion **58** (having a first D-shaped cross-section) is inserted into the bore **166** (having a second D-shaped cross-section) within the socket portion **150** of the body member **140** of the quick connect interface **14** until the bearing surface **90a** or **90b** having a shape conforming to a partial spiral around the longitudinal axis **18** of the replaceable tool **16** engages the alignment member **180** within the bore **166** causing rotation of the coupling end **52** of the replaceable tool **16** to align the flat portion **72** with the flat portion **170**. In some embodiments, the method further comprises moving the coupling sleeve **144** surrounding at least a portion of the body member **140** to an unlocked position prior to inserting the coupling portion **58** into the bore **166** and maintaining the coupling sleeve **144** in the unlocked position during the step of inserting the coupling portion **58** into the bore **166**. Once the coupling portion **58** is full inserted into the bore **166**, the coupling sleeve **144** is moved to the locked position to maintain a portion of the bearing **80** in the peripheral groove **76** thereby preventing removal of the coupling portion **58** from the bore **166**.

(52) To make the replaceable tool **16**, at least one bearing surface **90a** and/or **90b** having a partial spiral shape is formed on the coupling portion **58**. The coupling portion **58** has a D-shaped cross-section of the shaft **50** of the replaceable tool **16**. The shaft **50** having the tool end **54** opposite to the coupling end **52**, and the longitudinal axis **18** extending from the tool end **54** to the coupling end **52**. The coupling portion **58** extending from the coupling end **52** towards the tool end **54**.

(53) In some embodiments, the bearing surface **90** is a first bearing surface **90a**, and wherein the step of forming the first bearing surface **90a** is defined further as forming the bearing surface such that the partial spiral extends around the longitudinal axis **18** in a clockwise direction.

- (54) In some embodiments, the second bearing surface **90b** having a second shape conforming to a second partial spiral extending around the longitudinal axis **18** is formed and located at the coupling end **52** of the shaft **50**. The second bearing surface **90b** extends in a counter-clockwise direction.
- (55) It is discussed above that the quick connect interface **14** may be an AO quick connect interface. It should be understood that the techniques described herein may be readily adapted to work with pull or push variants of this type of interface. It should also be understood that at least some of the techniques may be readily adapted to work with other types of interfaces, for example, $\frac{1}{4}$ square, Zimmer, Hudson or other types of interfaces in which the coupling portion **58** of the shaft **50** does not include a D-shaped cross-section. It should also be understood that the various components of the quick connect interface **14** can be made utilizing various manufacturing techniques, such as 3D printing, milling, molding or the like, and then assembled. In some embodiments, the entire quick connect interface **14** can be made in an assembled form using a 3D printer and 3D printing techniques.
- (56) Similarly, while it is discussed above that one bearing **80** is used, it should be understood that a different number of bearings **80** can be used, such as only one, or two or more. Further, the bearing **80** need not be a ball bearing. For example, a cylindrical roller bearing may be substituted for the bearing **80** discussed above.
- (57) Further, while it is discussed above that the bias assembly **142** include a compression spring, it should be understood that other types of springs may be used for urging the coupling sleeve **144** to a locked position. For example, in some embodiments, one or more tension springs may be employed.
- (58) It should be understood that the quick connect interface **14** and the replaceable tool **16** may be constructed from a variety of materials, including stainless steel, titanium, aluminum, other metals, plastics, and combinations thereof.
- (59) The foregoing description provides illustration and description, but is not intended to be exhaustive or to limit the inventive concepts to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of the methodologies set forth in the present disclosure.
- (60) Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure includes each dependent claim in combination with every other claim in the claim set.
- (61) No element, act, or instruction used in the present application should be construed as critical or essential to the invention unless explicitly described as such outside of the preferred embodiment. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A replaceable tool, comprising: a shaft having a tool end, a coupling end opposite the tool end, and a longitudinal axis extending between the coupling end and the tool end, the shaft having: a tool portion defining the tool end; a coupling portion defining the coupling end, the coupling portion having an outer peripheral surface shaped into a flat portion coextensive with an arcuate portion cooperatively forming a D-shaped cross-section, the flat portion and the arcuate portion extending along at least a portion of the longitudinal axis; the outer peripheral surface defining a peripheral groove extending transversely with respect to the longitudinal axis; the peripheral groove configured to receive at least a portion of a bearing; and the outer peripheral surface also

defining a bearing surface having a shape conforming to a partial spiral extending around the longitudinal axis and located at the coupling end of the shaft.

2. The replaceable tool of claim 1, wherein the bearing surface has a first end and a second end opposed from the first end, the first end originating at the coupling end of the shaft and the second end terminating at the flat portion of the outer peripheral surface of the shaft.

3. The replaceable tool of claim 2, wherein the bearing surface extends at an angle between about 30 degrees and about 150 degrees relative to the longitudinal axis of the replaceable tool.

4. The replaceable tool of claim 1, wherein the bearing surface is a first bearing surface, wherein the shape is a first shape, and wherein the partial spiral is a first partial spiral, and further comprising a second bearing surface having a second shape conforming to a second partial spiral extending around the longitudinal axis and located at the coupling end of the shaft, the first bearing surface extending in a clockwise direction around the longitudinal axis, and the second bearing surface extending in a counter-clockwise direction around the longitudinal axis.

5. The replaceable tool of claim 4, wherein the coupling portion has a length, and wherein the first bearing surface and the second bearing surface meet within a distance of 0% to 10% of the length from the coupling end of the shaft.

6. The replaceable tool of claim 5, wherein the first bearing surface and the second bearing surface meet at a meeting location, the meeting location being within a range of plus 150 degrees from the flat portion to minus 150 degrees from the flat portion.

7. The replaceable tool of claim 5, wherein the flat portion has an end located a first distance from the coupling end, and wherein the first bearing surface and the second bearing surface meet within a second distance from the coupling end, the second distance being less than the first distance.

8. A quick connect interface, comprising: a body member having a socket portion and a shaft member, the body member including a first end and a second end opposite the first end, and a longitudinal axis extending through the body member from the first end to the second end, the shaft member extending from the socket portion; the socket portion having an outer surface and an inner surface, the socket portion having a front end and a rear end, the inner surface defining an opening at the front end, and defining a bore extending from the opening at the front end towards the rear end, the inner surface having a flat portion and an arcuate portion coextensive with the flat portion cooperatively forming a D-shaped cross-section; the socket portion having an alignment member within the bore, the alignment member having an outer surface defining at least a portion of the flat portion of the inner surface of the socket portion; the alignment member having a front surface facing the front end of the socket portion; and the front surface having at least one alignment surface extending at an angle between 30 degrees and 60 degrees relative to the longitudinal axis; the socket portion having a hole extending from the outer surface to the inner surface; a ball bearing positioned in the hole in the socket portion; a coupling sleeve extending around at least a portion of the socket portion, the coupling sleeve including a first portion and a second portion, the first portion being shaped so as to urge the ball bearing to partially extend through the hole into the bore of the socket portion, the second portion being shaped so as to not urge the ball bearing; and a bias assembly positioned adjacent to the socket portion and the coupling sleeve and operable to urge the coupling sleeve away from the rear end of the socket portion to a locked position.

9. The quick connect interface of claim 8, wherein the hole has a lip extending around a circumference of the hole at the inner surface, the ball bearing having a diameter greater than a smallest diameter of the lip at the inner surface so that the ball bearing may only partially cross a plane of the inner surface of the socket portion.

10. The quick connect interface of claim 9, wherein the sleeve is movable along the socket portion from an unlocked position to the locked position.

11. The quick connect interface of claim 9, wherein the socket portion has a linear groove in the outer surface defining an opening at the rear end and extending from the opening at the rear end towards the front end.

12. The quick connect interface of claim 11, wherein the coupling sleeve has a pin extending from an inner surface of the coupling sleeve into the linear groove.
 13. The quick connect interface of claim 9, wherein the bore has a diameter, and wherein the alignment member has a height in a range from 10% to 25% of the diameter of the bore.
 14. The quick connect interface of claim 9, wherein the at least one alignment surface is planar.
 15. The quick connect interface of claim 8, wherein the at least one alignment surface is a first alignment surface, and wherein the front surface of the alignment member further comprises a second alignment surface extending away from the first alignment surface, the first alignment surface and the second alignment surface forming a V-shape.
 16. The quick connect interface of claim 15, wherein the second alignment surface extends at an angle between 30 degrees and 60 degrees relative to the longitudinal axis.
 17. The quick connect interface of claim 15, wherein the first alignment surface and the second alignment surface are arranged to form an angle in a range from 60 degrees to 90 degrees.
 18. A method, comprising: inserting a coupling portion having a first D-shaped cross-section of a replaceable tool into a bore having a second D-shaped cross-section within a socket portion of a body member of a quick connect interface until a bearing surface having a shape conforming to a partial spiral around a longitudinal axis of the replaceable tool engages an alignment member within the bore causing rotation of the coupling end of the replaceable tool.
 19. The method of claim 18, further comprising moving a coupling sleeve surrounding at least a portion of the body member to an unlocked position prior to inserting the coupling portion into the bore and maintaining the coupling sleeve in the unlocked position during the step of inserting the coupling portion into the bore.
 20. The method of claim 19, further comprising moving the coupling sleeve to a locked position after the coupling portion is inserted into the bore.
 21. A method of making a replaceable tool, comprising: forming a bearing surface having a partial spiral shape on a coupling portion having a D-shaped cross-section of a shaft of the replaceable tool, the shaft having a tool end opposite to a coupling end, and a longitudinal axis extending from the tool end to the coupling end, the coupling portion extending from the coupling end towards the tool end.
 22. The method of claim 21, wherein the bearing surface is a first bearing surface, and wherein the step of forming the first bearing surface is defined further as forming the bearing surface such that the partial spiral extends around the longitudinal axis in a clockwise direction.
 23. The method of claim 22, further comprising forming a second bearing surface having a second shape conforming to a second partial spiral extending around the longitudinal axis and located at the coupling end of the shaft, the second bearing surface extending in a counter-clockwise direction.
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